Faithful per-layer geometric readout of the ESMFold trunk, and the energy-manifold adherence it reveals

William Klett
Princeton University

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Abstract

We introduce a *faithful per-layer geometric readout* of the ESMFold trunk: a distogram autoencoder trained, at each block, to reconstruct that block's pair representation z_t, with cross-layer consistency chaining the readout back to the model's own output head at the in-distribution final layer—so that no native structure ever enters the loss. Unlike the model's own head applied out-of-distribution to intermediate layers, and unlike ground-truth-supervised per-block decoders, this readout measures *what the trunk believes* rather than *what is decodable from it*. Using it, we find that the trunk's intermediate geometric beliefs adhere to a low-energy manifold: the trajectory moves through clash-free, increasingly well-packed configurations and avoids high-energy geometry even on proteins the model ultimately folds incorrectly, whereas the straight path between the same two waypoints tunnels through sterically forbidden configurations (atomic overlap, energy spiking by ∼8 orders of magnitude). This adherence is *emergent*—the trunk is trained on endpoints, not trajectories—and it is *not* first-order gradient descent on energy. The result is, to our knowledge, the first reading of *what the trunk believes* (as opposed to what an external instrument can extract from it), and it supports a mechanistic account of single-pass folding as manifold-adherent reorganization rather than energy gradient descent.

1 Introduction

The gap. Single-pass structure predictors such as ESMFold are accurate but mechanistically opaque. The trunk iteratively refines a pair representation $z \in \mathbb{R}^{L\times L\times 128}$ over ∼48 blocks [CHECK: confirm exact block count for the cached model (47/48)] and emits a single prediction with no recycling and no MSA. Reading the trunk's *intermediate* states faithfully would show *how* a structure assembles, not merely *that* it does.

Two questions, one of them open. Interpretability of these models divides into two questions: *what the model is thinking about*—the content that is decodable from a representation (linear probes, sparse autoencoders)—and *what those thoughts are*—the model's own evolving belief. The first is well studied; the second is not, because the obvious instruments for it are confounded (§2). This paper answers the second, faithfully, for the trunk's geometric belief.

Contributions.

1. **A faithful per-layer geometric readout.** A distogram autoencoder probe per block, trained on reconstruction of z_t plus adjacent-layer consistency, anchored only at the in-distribution

final layer to the model's own head—no native structure in the loss (§2.3, §2.4). We give the faithfulness criterion it is built on and validate it (§2.5). [NEXT-PAPER: resolve the readout's name vs. the prior "PSA" (physical structure autoencoder) paper to avoid collision.]

2. **The finding it enables.** The trunk's intermediate beliefs trace a trajectory that adheres to the low-energy manifold even on incorrect folds, while a direct path between the same endpoints tunnels through forbidden high-energy geometry (§3). The adherence is emergent (endpoint-only training) and is not gradient descent.

Roadmap. Section 2 situates the method against prior interpretability work and derives the faithfulness criterion. Section 3 presents the energy-manifold results. Sections 4–6 give bounds, directions, and a closing synthesis.

2 Prior work and what faithfulness requires

2.1 Two kinds of interpretability question

What the model is thinking about (decodable content). Linear probes and sparse autoencoders treat the representation as something to extract from—an external instrument measuring what is recoverable.

- Sparse autoencoders: InterPLM [Simon & Zou]; **(author?)** [Adams et al.]; the Reticular Matryoshka-SAE on ESM2-3B [Reticular]. [VERIFY: each does what is attributed; read methods, not abstracts.]
- Linear decoders: **(author?)** [Lu et al. 2026] (phase diagram of distance decodability by depth); AlphaInterp on AlphaFold 3 [AlphaInterp]. [VERIFY: attribution.]

What those thoughts are (the model's evolving belief). The natural move is to apply the model's own output interface to an intermediate state—but this has two problems.

2.2 The second camp's two problems

Problem 1: the distogram head is out of distribution on intermediate layers. ESMFold's distogram head is trained on the distribution of *final-layer* representations. Applied mid-trunk it is out of distribution, and it shows it: the implied distance maps violate the triangle inequality across a wide middle band of layers (the "M-curve," Fig. 1). This is a contradiction in the field's joint commitments: geometry is reported to be *linearly decodable* from z into Euclidean space [Lu et al. 2026] [VERIFY: that linear-decodability is attributable to Lu et al.], yet the head's intermediate readout is *non-metric*. Both cannot describe the same representation. The resolution: the non-metricity belongs to the *instrument*, not to z.

Problem 2: ground-truth-supervised per-block readouts measure the wrong thing. AlphaFold retrains a structure module per block against native structure [Jumper et al. 2021] [VERIFY: native-supervised per-block readout, Supp. Methods 1.14]. This avoids the OOD problem but measures *downstream sufficiency*, not *content*: how recoverable the final answer is from layer t, not what layer t represents.

The proof analogy. One could train a model to complete a proof from any single line; its success rate measures how *determined* the conclusion is by that line, not what the line *says*. A native-supervised per-block readout reads the proof's completability, not the content of the line. The consequence matters here: a confidently *wrong* intermediate belief has definite content—a specific

wrong geometry—but a ground-truth decoder can only report low decodability and will miss that a coherent wrong belief is present. (This is exactly the regime our false-fold results in §3.4 probe.)

2.3 What faithfulness requires

We define the trunk's geometric belief at layer t as the distance distribution that, among all distance distributions, best *reconstructs* z_t.

- **Reconstruction fidelity to z_t, not predictive fidelity to native.** Reading content, not implication.
- **Restriction to distance distributions.** Reading the *geometric* belief specifically, projecting out sequence/coevolutionary content.

Reconstruction is the right content criterion for a causal reason: z_t is what the downstream trunk actually consumes, so reconstruction-recoverable content is exactly the content that flows forward. The probe reads what the network reads; a native decoder reads what an external instrument can extract, which need not coincide.

2.4 Probe construction

At each block t we train a per-pair bottleneck, `PairDistogramBottleneck` (`pair_dim`= 128, `bins`= 64, `hidden_dim`= 256), that maps a single pair vector $z_t[i,j]$ to a 64-bin distogram over the same distance grid ESMFold's own head uses ($\approx$ 2.3–21.7 Å), then decodes back to $z_t[i,j]$. The loss has exactly two per-pair terms:

$$\mathcal{L}_t \;=\; \underbrace{\left\| \,\mathrm{dec}(\mathrm{enc}(z_t[i,j])) - z_t[i,j] \,\right\|^2}_{\text{reconstruction of } z_t} \;+\; \lambda \underbrace{\mathrm{KL}\big(\mathrm{enc}(z_t[i,j]) \,\big\|\, p_t^{\mathrm{tgt}}[i,j]\big)}_{\text{adjacent-layer consistency}}. \tag{1}$$

Linking-back anchoring. The layer-46 probe's consistency target p_{46}^{tgt} is ESMFold's own distogram head applied to z_{47}—the head used *in distribution*, at the layer it was trained for. Each earlier probe is anchored to the just-trained next probe: $p_t^{\mathrm{tgt}} = \mathrm{enc}_{t+1}(z_{t+1})$. The chain is anchored only at the final, in-distribution layer; no native structure enters at any point.

Iterative-refinement assumption (imported). Reading distances block by block presupposes that geometric content changes *gradually* across blocks, so each backward link is a small extrapolation from a good anchor. We state this as an assumption, supported by prior evidence that distance/geometric content accumulates progressively through the trunk [Lu et al. 2026] [VERIFY: phase-diagram / gradual-accumulation attribution]; we do not prove it here.

2.5 Faithfulness validation: the changes are geometric

The core argument. Reconstruction forces each probe to encode whatever distinguishes its layer. That information is either geometric or non-geometric—two cases with opposite signatures.

Case 1 (non-geometric). Forcing layer-distinguishing information through a per-pair *distance* code mis-encodes it. Because probes are chained, the Euclidean fraction would decay multiplicatively with distance from the anchor: embedding stress would rise monotonically toward early layers and triangle violations would accumulate.

Case 2 (geometric). Independent per-pair readouts come out mutually metric-consistent with *no* enforcement: stress stays flat and low, and violations stay at zero at all depths.

[FIGURE TO GENERATE]

Paired triangle-inequality violation rate across trunk depth: ESMFold's own head applied to intermediate z_t (M-shaped, violations across the middle band) vs. the faithful probe (flat at zero, all depths). Generate with `scripts/eval_triangle_violations.py` extended to also evaluate the native head per layer; `scripts/plot_triangle_inequality.py` for the figure.

Figure 1: Methodological centerpiece. The head is non-metric mid-trunk; the probe, with the same freedom, is metric-consistent at every depth—so the geometry is a property of z, read faithfully by the probe, not an artifact of the readout.

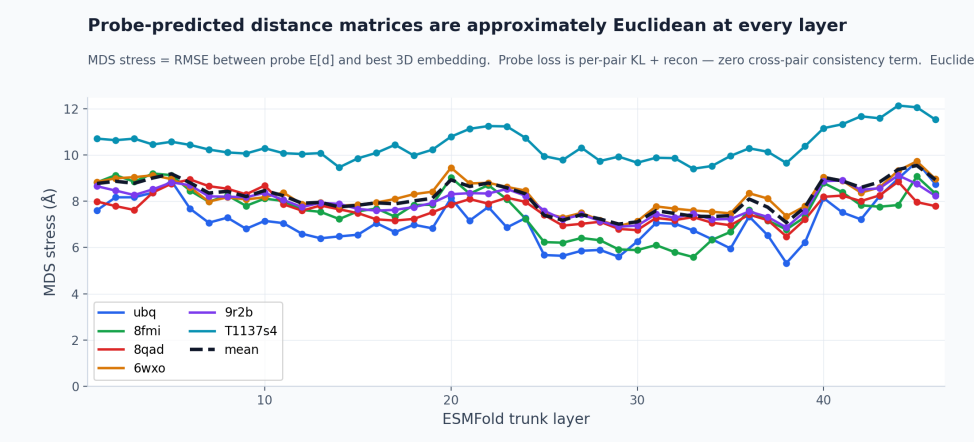


Figure 2: MDS embedding stress of the probe-read distance matrix across all 46 trunk layers, for six benchmark proteins. Stress is flat and low at every depth—no growth toward early layers—consistent with Case 2: the probe reads coherent geometry rather than mis-encoding non-geometric content. [CHECK: add the native-structure embedding-error reference line.]

We observe Case 2. Across 57 proteins (good and bad folds) and all 46 layers, sampling random triples (i, j, k) and checking $d(i, k) > d(i, j)+d(j, k)$, the violation rate is 0.0000 (∼8M triple-checks). [CHECK: re-confirm the exact triple count and that the rate is identically zero at every depth, not just on average.] Embedding stress is flat across depth and low—comparable to the embedding error of the native structures themselves (∼5–12 Å, Fig. 2)— even for the protein whose final TM-score is near zero. [CHECK: report the stress band precisely and the per-layer reconstruction R^2; confirm no early-layer reconstruction degradation.]

The clincher. Nothing in the loss enforces Euclidean structure: no cross-pair coupling, no embedding term. The probe is *free* to be non-metric—and the model's own head, with the identical freedom, *is* non-metric on the same layers (Fig. 1). The probe is not. Unforced metric-consistency of independent per-pair readouts is possible only if z encodes coherent geometry for them to read. This is the methodological centerpiece: the contrast between the head's M-shaped violation curve and the probe's flat-zero curve.

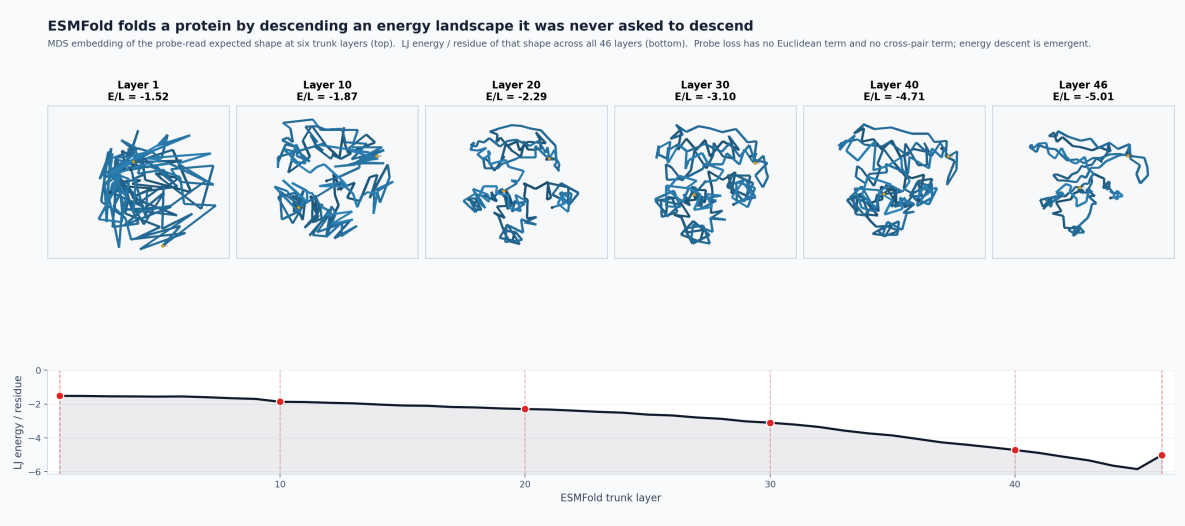


Figure 3: Keyframe montage of one protein's probe-read intermediate structures (fixed camera, aligned), with the per-layer energy curve beneath. The structure self-assembles through large, coordinated motions; the energy descends in lockstep. Motivation for the quantitative tests in §3.2–§3.6.

3 The trunk's beliefs adhere to the low-energy manifold

Topic sentence. The trunk's intermediate structures trace a trajectory that adheres to the low-energy manifold, even when the model folds incorrectly.

3.1 The observation that prompted the question (motivation, not evidence)

Watching the MDS-embedded intermediate structures evolve (fixed camera, Kabsch-aligned to the final layer), the trunk makes large, *coherent* global reconfigurations—and does so even on proteins it ultimately folds incorrectly. Two things are apparent: the moves are large, and the moves are clean (coordinated, not thrashing). The puzzle: a large reconfiguration generically passes through bad geometry (clashes, high energy), yet the trunk's moves appear not to. Does the trajectory respect the energy landscape even during large moves, even when folding wrong? We frame this as motivation; the quantitative tests follow. The claim is "large coherent moves even when wrong," not "wrong folds move more"—the latter is unneeded.

3.2 Energy descends along the path—and the descent is packing, not clash-avoidance

We compute the Lennard-Jones energy of the probe's expected shape per layer,

$$E_t \;=\; \frac{1}{L}\sum_{|i-j|>3} 4\varepsilon\left[\left(\frac{\sigma}{\mathbb{E}[d_t(i,j)]}\right)^{12} - \left(\frac{\sigma}{\mathbb{E}[d_t(i,j)]}\right)^{6}\right], \qquad \sigma = 6.8\,\text{Å},\ \varepsilon = 1, \tag{2}$$

and observe a near-universal monotone decrease through the trunk (Fig. 4).

Two distinct energy phenomena, kept separate.

(i) *Path-level steric adherence* (§3.5): the trunk's *route* stays in the non-overlapping region; driven by the repulsive r^{-12} wall; dramatic magnitude (clashes $\to 10^3+$); a geometric / excluded-volume claim.

(ii) *Within-manifold packing descent* (this subsection and §3.3–§3.4): among already-valid, clash-free configurations, the trunk moves toward more favorably-packed ones; driven by the attractive/soft part of the potential; modest magnitude.

These do different jobs (boundary adherence vs. interior optimization); neither reduces to the other.

The descent is not the repulsive term—an argument from magnitude. The monotone decrease through the trunk is of order ∼5–10 per residue. Steric clashes under the *same* potential produce changes of order 10^3 and above (the chord spike, §3.5), because r^{-12} blows up on overlap. The repulsive wall is effectively a step—clashing (10^3+) or not (≈ 0)—with no regime that contributes a smooth change of size 5–10. Therefore a smooth descent of order 5–10 is far too small to be escape from the repulsive wall; it must reflect refinement in the soft, attractive, clash-free regime: the trunk improving favorable packing among configurations that are already physically valid. A second angle gives the same conclusion: the minimum interatomic distance stays comfortably above contact distance *throughout* the descent phase—there is no clash to escape while energy is descending. [CHECK: report min-interatomic-distance during the descent phase; available from the §3.5 analysis.]
Why it matters: a model that merely avoided clashes would be unremarkable; a model that smoothly, monotonically improves packing favorability *among valid structures*—even on wrong folds (§3.4)—is doing something optimization-like. The magnitude argument licenses calling the descent optimization rather than triviality. The immediate deflation to rule out next: is this just convergence toward the (low-energy) answer?

3.3 The packing descent is not just convergence (within-protein)

Deflation. Is the packing descent merely convergence toward the low-energy answer? A byproduct reading predicts that the rate of geometric approach to the destination and the rate of energy descent are positively coupled.

Result. Within a single protein, across layers, the layers of fastest *geometric convergence* are *not* the layers of fastest *energy descent*; the two rates are anti-correlated. [CHECK: confirm the within-protein (not cross-protein) version is computed—this avoids the initial-condition / difficulty confound entirely.] This temporally separates the two processes within one computation: energy descent is distinct from geometric approach, evidence for *staged* computation rather than a shadow of convergence. It connects to the onset of energy-gradient alignment around block 17 [CHECK: confirm onset block].

3.4 It is not just correctness (false-fold dissociation)

Deflation. Maybe energy simply tracks correctness—moving toward the right (low-energy) answer.
Result. On incorrect folds the trajectory still respects energy and avoids spikes, even though the destination is wrong. Successes and failures share the *procedure*; energy-avoidance is a property of how the trunk moves, not of where it lands. The wrong-fold cases are the natural experiment dissociating energy-cleanliness from answer-correctness, and they are exactly the regime a ground-truth-supervised decoder (§2.2) is blind to.

3.5 The path is special: steric manifold adherence vs. the chord

This is the path-level steric claim—phenomenon (i)—distinct from the packing descent. Between two of the trunk's own waypoints we compare the trunk's actual path against a straight-line interpolation, *in 3D coordinates* (not distance matrices: every intermediate is then a real conformation), with Kabsch-aligned endpoints so chord clashes are not a rotation-collapse artifact. [CHECK: endpoints Kabsch-aligned (done); use fine interpolation sampling so the spike is not a sampling coincidence.]

Why coordinates, not distance matrices. A linearly interpolated distance matrix can violate the triangle inequality and correspond to *no* structure; a linearly interpolated coordinate set is always a real arrangement of points. More decisively, distance interpolation *cannot* exhibit the effect: each interpolated pair distance lies between its two endpoint values, and LJ attains its maximum on any interval at an endpoint, so the distance-space chord can never rise above the endpoint envelope (we verify it stays low). A clash is intrinsically a Cartesian event—two residues far apart in both endpoints whose straight-line position paths cross—so only coordinate interpolation can reveal it. The 3D embedding is obtained by classical MDS refined with SMACOF stress majorization, which is required so that interpolant clashes are genuine geometry and not embedding collapse.

ubq result. For ubiquitin the chord drives atoms through one another: minimum interatomic distance $\to 0$ and LJ energy spikes by $\sim$8 orders of magnitude at the midpoint, while the trunk's own path maintains physical separation and descends monotonically (Fig. 6). The endpoints (blocks 20 and 46) differ by an RMSD of 11.6 Å—larger than the protein's own radius of gyration (10.9 Å)—so this is a genuine global rearrangement, and the embedding is faithful (stress 1.3 Å).

The claim is geometric, not energetic. The headline metric is potential-*free*: the minimum interatomic distance (equivalently, the fraction of pairs in steric clash) along each path. The chord reaches atomic overlap (physically impossible); the trunk stays above contact distance. The LJ spike is merely the overlap rendered as a number. We therefore do *not* run a multi-potential robustness sweep: LJ, soft-sphere, DFIRE, etc. share the same short-range repulsive core, so they are not independent tests—they all spike on overlap by construction. The choice of potential does not matter *because* the claim is excluded volume, which holds independent of any energy model. This converts the "you only used one potential" objection into a non-issue.

Endpoint separation explains the null cases. The effect appears precisely when the two on-manifold endpoints genuinely differ (Table 1). For 8fmi, blocks 20 and 46 are 1.0 Å apart—8% of its radius of gyration, the refinement regime—so there is nothing to morph through and the chord is correctly flat. ubq's endpoints differ by more than its own size, and the chord spikes cleanly. T1137s4 is intermediate on both axes. So ubq is the case that actually exercises the test, not a cherry-pick: the chord spike scales with how distinct the endpoints are.
Claim earned: the trunk navigates along the physically realizable region rather than tunneling through forbidden geometry—not any path to the destination works.

3.6 It is not gradient descent; it is emergent

Is the trunk descending energy by gradient steps? We measure $\cos(\Delta z_t, -\nabla_z E)$.

Table 1: Endpoint separation (blocks 20→46) and chord behavior. The spike appears exactly when the two on-manifold states genuinely differ.

protein	endpoint RMSD (Å)	RMSD / R_g	embed. stress (Å)	chord
8fmi	1.01	0.08	3.4 / 2.6	flat (endpoints near-identical)
T1137s4	8.72	0.66	5.0 / 5.2	spikes; embedding-noise-limited
ubq	11.56	1.06	2.6 / 1.3	spikes ∼8 OOM; clean

The alignment is real but far below gradient descent. In $D \approx 7.4 \times 10^5$ dimensions [CHECK: $D = L^2 \cdot 128$; state L used], random directions are orthogonal to within $\pm 1/\sqrt{D} \approx 0.001$, so a measured cosine of ∼0.05 is a $\sim 43\sigma$ departure from chance ($p \sim 10^{-401}$)—the energy-aligned component is real beyond any doubt (random baseline $\sim 10^{-4}$, Fig. 8). [CHECK: column meaning in `cosine_alignment_summary.csv`: confirm the energy-gradient column and the representative value; resolve the earlier −0.01 vs. 0.05 discrepancy and restate the GD-ruling-out argument consistently.] But 0.05 is far below the ≈ 1.0 that literal first-order gradient descent would produce. Moreover, isotropic noise can only *attenuate* a measured cosine toward the orthogonal null, so 0.05 is a *lower bound* on the true alignment, not an estimate.[1]

Synthesis. The trunk does *not* perform gradient descent on energy, yet its updates carry a real, lower-bounded energy-aligned component, its path adheres to the physically realizable region (§3.5), and it improves packing favorability monotonically within that region (§3.2–§3.4). It implements an amortized procedure that *arrives at* well-packed, sterically valid configurations without *descending toward* them by gradient steps.

Emergence (keystone). The trunk is trained on endpoints, not trajectories—it was never shown a folding path. Both the steric path-adherence and the packing descent are therefore properties of the learned procedure, not trained objectives. This closes the deflationary readings that require energy to be present in the training signal.

4 Limitations

- **Energy claims are split and neither rests on a chosen potential.** The steric / path-adherence claim (§3.5) is geometric—excluded volume, reported as minimum interatomic distance—hence potential-free. The packing-descent claim (§3.2–§3.4) is established as attractive-regime by the magnitude argument (descent ∼5–10 vs. clash scale $\geq 10^3$ under the *same* potential), not by the LJ form. We deliberately do not claim potential-independence via a multi-potential sweep, since such potentials share a repulsive core and are not independent tests. Both claims concern coarse-grained Cα geometry, not full-atom energetics.
- **Cα-only geometry** via MDS embedding of probe distances; side chains, backbone torsions, and full clash terms are not captured.
- **Faithfulness assumptions.** The readout rests on the gradual-geometric-change assumption (imported, supported by **(author?)** [Lu et al. 2026], not proven here) and on the anchor being in-distribution at the final layer.

[1] A clean direction diluted by per-coordinate noise of magnitude ∼0.023 yields a measured cosine near 0.05; detectability ≠ magnitude.

- **Scope of content.** The probe reads geometric content specifically; non-geometric content in z is outside the method's scope. [NEXT-PAPER: the trunk carries information beyond the geometric content the probe reads; characterizing it is a direction we are pursuing.]
- **Mechanism characterized negatively.** We show not-first-order-GD and emergent; the positive identity of the amortized procedure is not established.
- **Scale / generality.** $N =$[CHECK: number of proteins] proteins, ESMFold specifically; generality beyond ESMFold is unverified here.

5 Future work

- **Identify the amortized procedure** positively (proximal step? preconditioned / natural gradient? alternating projection?), following directly from the negative characterization in §3.6.
- **Causal intervention via the faithful probe:** the repair experiment—inject a high-energy perturbation and test whether the trunk restores manifold adherence—the interventional version of §3.5's observational result.
- **Non-geometric content** the probe does not capture, and its role in the computation. [NEXT-PAPER: stake priority only; do not reveal the reliability metric.]

6 Conclusion

We presented a faithful per-layer geometric readout of the ESMFold trunk—content via reconstruction of z_t, not decodability via supervision; anchored in-distribution at the final layer, with no native structure in the loss—and used it to show that the trunk's intermediate beliefs adhere to the low-energy manifold even when the model folds incorrectly. The adherence is emergent from endpoint-only training and is not gradient descent: the trunk implements an amortized procedure that arrives at well-packed, sterically valid configurations without descending toward them by gradient steps, while a straight path between the same waypoints tunnels through forbidden geometry. This is a reading of what the trunk *believes*, not merely what is decodable from it, and it supports a mechanistic account of single-pass folding as manifold-adherent reorganization.

References

[Adams et al.] Adams et al. *Sparse autoencoders for protein language models.* [VERIFY: full citation + claim.]

[AlphaInterp] AlphaInterp: interpretability for AlphaFold 3. [VERIFY: authors, venue, claim.]

[Jumper et al. 2021] Jumper et al. *Highly accurate protein structure prediction with AlphaFold.* Nature, 2021. [VERIFY: per-block structure-module supervision, Supp. Methods 1.14.]

[Lin et al. 2023] Lin et al. *Evolutionary-scale prediction of atomic-level protein structure with a language model (ESMFold).* Science, 2023.

[Lu et al. 2026] Lu et al. *Phase diagram of distance decodability across depth.* 2026. [VERIFY: linear decodability + gradual accumulation attribution.]

[Reticular] Reticular. *Matryoshka sparse autoencoders on ESM2-3B.* [VERIFY: citation + claim.]

[Simon & Zou] Simon & Zou. *InterPLM: interpreting protein language models with sparse autoencoders.* [VERIFY: citation + claim.]



Figure 4: Per-protein LJ energy of the probe's expected shape across the trunk, arranged by fold quality. Descent is near-universal; deeper, smoother descents belong to better folds. The decrease is of order 5–10/residue—far below the 10^3+ scale of steric clashes (§3.5)—so it reflects packing refinement within the clash-free regime, not escape from overlap.

[FIGURE TO GENERATE]

Within-protein, per-layer geometric-convergence rate vs. energy-descent rate; the two are anti-correlated. Generate with `scripts/analyze_convergence_vs_energy.py` (within-protein variant).

Figure 5: Energy descent and geometric approach peak at different layers within a single computation—staged, not a byproduct of convergence.

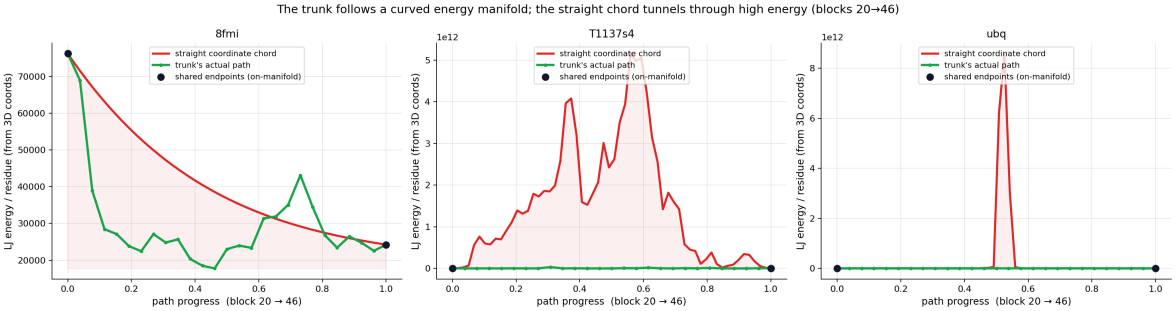


Figure 6: Trunk path (green) vs. straight coordinate chord (red) between two on-manifold waypoints, energy from realized 3D coordinates. The chord tunnels through high-energy, sterically overlapping geometry; the trunk path stays low. [CHECK: regenerate as a clean ubq-only panel with log y-axis; the magnitude is a hard-LJ near-collision artifact, the honest claim is the qualitative separation.]

[FIGURE TO GENERATE]

Four-distribution comparison of the potential-free headline metric (minimum interatomic distance / steric-clash fraction): trunk-success, trunk-failure, chord-success, chord-failure. The two trunk distributions overlap and stay safe; both chord distributions hit atomic overlap. Generate by extending `scripts/manifold_chord_experiment.py` to emit per-path min-distance and clash-fraction over a success/failure protein set.

Figure 7: Trunk-path steric adherence is the same on failures as on successes; both chords tunnel. The trunk navigates *along* the realizable region rather than through forbidden geometry.

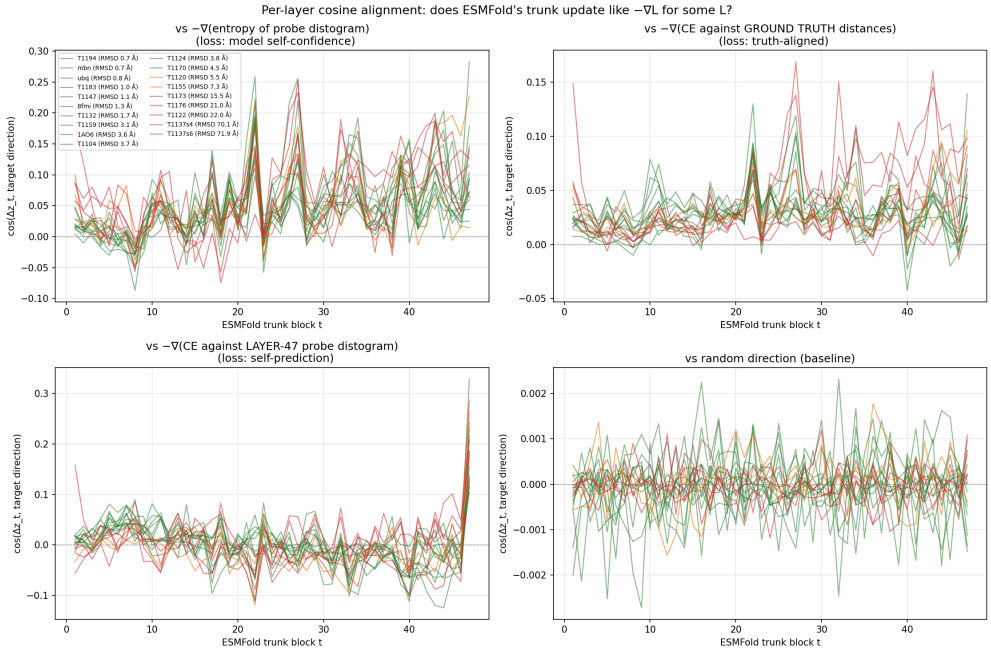


Figure 8: $\cos(\Delta z_t, -\nabla_z E)$ across layers and proteins. The alignment is small but vastly above the random-direction baseline ($\sim 10^{-4}$) and far below the ≈ 1.0 of literal gradient descent; with isotropic noise it is a lower bound on the true alignment.